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Self-driven tandem alcoholysis for full upcycling of waste polycarbonate plastics

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Abstract

Conventional alcoholysis processes for polycarbonate (PC) recycling require excess methanol input, high reaction temperatures and complicated downstream product separation steps, severely limiting environmental and economic sustainability. Here we report a facile, self-driven tandem alcoholysis process to address these challenges. Methanol-based PC alcoholysis reaction is coupled to a downstream glycerol transesterification step to rapidly convert the dimethyl carbonate intermediate into methanol, thus forming a closed-loop methanol flow to boost the overall reactions. Additionally, the reactions are catalyzed by eggshell-derived multimetallic CaO, which provides abundant acid–base sites for synergistically deprotonating alcohols and activating ester bonds to facilitate a selective conversion. This integrated design not only enables 98.8% less methanol usage and significantly lower reaction temperature (80 versus >130 °C) than the PC-alone alcoholysis process, but also generates bisphenol A (96% yield) and 5-hydroxymethyl-1,3-dioxolan-2-one (90% yield) as higher-value products. The process is adaptable to treating mixed-plastic feeds with robust performance. The pilot-scale trials (>5 kg) using real mixed plastic waste demonstrate superior upcycling performance and distinctly improved environmental and economic benefits over the existing industrial synthesis routes. Overall, this scalable, self-driven tandem alcoholysis process offers a sustainable, industrially-viable platform for upcycling PC-containing waste plastics.

Introduction

Polycarbonate (PC) is one of the most widely used engineering plastics worldwide, and approximately 6 million tons of PC waste are generated every year globally.^{1,2} These wastes are persistent in the environment and their decomposition gradually releases hazardous microplastics and toxic chemicals, raising widespread environmental and health concerns.³⁻⁹ On the other hand, the conventional chemical industry based on fossil fuels is environmentally detrimental and unsustainable. Upcycling such waste plastics into valuable chemicals is considered a promising strategy to address both challenges.¹⁰⁻¹² The currently prevailing approaches for chemical upcycling of PC plastics include pyrolysis,¹³ hydrolysis,¹⁴ and alcoholysis¹⁵ (Supplementary Fig.1). Among these, alcoholysis is known to yield valuable products including BPA and dimethyl carbonate (DMC) from PC conversion at relatively mild operating conditions.¹⁵⁻¹⁷ Nevertheless, the reversible and thermodynamically unfavorable nature of PC alcoholysis reaction limits the conversion kinetics. Thus, excess MeOH dosage and relatively high temperatures (>130 °C) are always needed for this reaction, raising the energy consumption and process complexity and resulting in considerable safety risks. Moreover, the co-existing polyesters (e.g., PET) and common additives are readily susceptible to thermal decomposition, generating undesirable by-products and lowering the product purity. The relatively low market value of DMC product in such processes also limits the economic incentive for their practical application.¹⁸

Here, we developed a self-sustaining tandem alcoholysis process to address above challenges of PC upcycling. In this process, the PC alcoholysis reaction is fundamentally reconfigured by introducing glycerol as a co-substrate in one-pot reaction system. Specifically, the DMC derived from PC alcoholysis is immediately consumed by glycerol via transesterification reaction, yielding hydroxymethyl dioxolanone (HMDL, a high-value platform chemical) and simultaneously regenerating MeOH. To support the tandem reactions, eggshell-derived bio-CaO with multimetallic oxides was introduced as a low-cost catalyst, which provides abundant basic and Lewis acid sites to favor an efficient and selective PC conversion. In this way, a MeOH closed-loop is established to drive both the PC alcoholysis and glycerol transesterification reactions, enabling a rapid and near-complete conversion of PC and glycerol under unprecedentedly mild conditions and without requiring complex separation processes (Fig.1a; Supplementary Fig.2). Therefore, such a mutually-reinforcing tandem alcoholysis process represents a significant advance over the existing processes for PC and glycerol upcycling.

Benefited from the self-driven tandem alcoholysis design, the overall process achieved efficient PC upcycling at much lower temperature (80 vs. >130 °C) and with 97–99% less MeOH usage than

conventional alcoholysis processes, substantially improving process safety and environmental benefits while reducing operating costs. Meanwhile, the generation of HMDL product, with nearly tenfold higher market value than DMC, further improves its economic viability. Importantly, the facile operation and mild reaction conditions make the tandem alcoholysis process readily scalable and suitable for practical applications. The thermal decomposition of common additives and co-existing polymers (e.g., PE, PP and PET) in real-world mixed plastic waste was avoided, drastically improving the product quality and reducing secondary pollution. Additionally, the process scalability was demonstrated in a pilot-scale setup with a simple glass reactor for processing >5 kg mixed plastic waste. Life cycle assessment (LCA) and techno-economic analysis (TEA) confirm a significantly reduced environmental impact and distinct economic benefits over the prevailing industrial routes for BPA and HMDL production, highlighting its strong potential for practical industrial deployment and commercial application.

Results

Catalyst selection for tandem alcoholysis of PC and glycerol

The process flow of PC tandem alcoholysis, involving alcoholysis-transesterification cascade reactions, is illustrated in Fig. 1a and Supplementary Fig. 2. PC alcoholysis reactions typically involve basic reagents¹⁹ (e.g., NaOH, KOH) or catalysts such as metal oxides and metal salts^{20,21}. Here, base metal oxides were considered due to their superior catalytic activity for both alcoholysis and transesterification reactions and good reusability. We first screened several commercially available, low-cost metal oxides, including calcium oxide (CaO), magnesium oxide (MgO), and zinc oxide (ZnO). The abundant basic sites (e.g., OH⁻, O²⁻) and Lewis acidic sites (e.g., Zn²⁺) of these catalysts are expected to synergistically boost the tandem reactions: basic sites catalyze alcohol deprotonation to generate nucleophilic alkoxides (RO⁻), while the Lewis acid sites activate the carbonyl group of PC and glycerol to facilitate its nucleophilic attack by alkoxide ions (RO⁻)²²⁻²⁴. Our results show that, among all the tested base metal oxides, CaO exhibited the highest activity for the tandem alcoholysis reactions, achieving 81% PC conversion within 4 hours (Supplementary Fig. 3a-b).

Considering that eggshell waste also contains abundant Ca and aiming to further reduce the operating cost, we explored the feasibility of using an eggshell-derived material (named bio-CaO due to the formation of CaO after calcination) as an alternative catalyst for the tandem alcoholysis process. Interestingly, the bio-CaO exhibited even higher activity than the commercial CaO, achieving >99% PC conversion within 4 hours accompanied by efficient production of BPA (96% yield) and HMDL (90%

yield). Here, the further raised activity of bio-CaO over commercial CaO should be mainly attributed to its trace MgO and ZnO impurities, which serve as extra catalytic sites for the tandem reactions (detailed below). Therefore, the bio-CaO catalyst was adopted in subsequent experiments.

Reaction routes of the tandem alcoholysis process

In the PC tandem alcoholysis process, MeOH and glycerol were both added to boost PC conversion. A comparison of the reaction kinetics (Fig.1b, Supplementary Fig. 4) showed two-order-of-magnitude higher PC conversion rate (apparent rate constant $k = 0.91 \text{ h}^{-1}$) in the tandem reaction system than in the individual MeOH ($k = 0.013 \text{ h}^{-1}$) and glycerol groups ($k = 0.010 \text{ h}^{-1}$), confirming the essential roles of both MeOH and glycerol in driving the tandem reactions. The temperature-dependent PC conversion rate in the tandem alcoholysis system (Fig. 1c, Supplementary Fig. 5) and the considerable activation energy ($E_a = 97.99 \text{ kJ/mol}$) for PC alcoholysis (Fig. 1d) confirm its thermochemical reaction nature. Additionally, the fitting model reveals a relatively large pre-exponential factor ($A = 1.16 \times 10^{14} \text{ h}^{-1}$), indicative of a high probability of molecular collisions at the catalyst surface, supporting a critical catalytic role of bio-CaO in the tandem alcoholysis reactions.^{25, 26}

To clarify how MeOH and glycerol are involved in the alcoholysis and transesterification reaction steps respectively, we next examined the PC depolymerization kinetics and product yields at 80 °C operation under different scenarios of MeOH and glycerol addition. Nearly-complete PC conversion with persistently minimal DMC accumulation (<5%) was achieved upon simultaneous addition of MeOH and glycerol (Fig. 1e). In contrast, the PC depolymerization and BPA production were drastically suppressed in the MeOH-only group (initially free of glycerol), and DMC instead of HMDL was obtained as the major product. Notably, upon subsequent glycerol injection, the reaction kinetic was immediately restored to a comparable level to that in the tandem alcoholysis system and HMDL production was initiated (Fig.1f). These results confirm that glycerol is essential for the transesterification reaction, converting DMC into HMDL and simultaneously regenerating MeOH to drive the overall tandem reactions. A similar phenomenon was observed in a glycerol and MeOH sequential addition scenario (Fig. 1g), where negligible PC conversion occurred in the presence of glycerol alone, further supporting the essential role of MeOH in initiating PC alcoholysis while excluding the direct involvement of glycerol in PC conversion (Supplementary Fig. 6).

The above tandem reactions were further verified by in-situ attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy. During PC alcoholysis with MeOH alone (Fig. 1h), the FTIR $\nu_{\text{as}}(\text{OH})$ signal at 3300 cm^{-1} gradually decreased, corresponding to the consumption of hydroxyl

group in MeOH²⁷. As the reaction proceeded, the hydroxyl band at 1040 cm⁻¹ gradually evolved into triply bridged hydroxyl groups (1030, 1045, and 1056 cm⁻¹)^{28, 29}, while the signals of β -interaction (1056 cm⁻¹) and γ -interaction (1030 cm⁻¹) were progressively intensified, corresponding to the process of MeOH hydroxyl group deprotonation at the bio-CaO surface to form alkoxy bonds³⁰. In contrast, upon simultaneous addition of MeOH and glycerol to initiate tandem alcoholysis (Fig. 1i), the FTIR signal of O–H stretching initially decreased and then slightly rebounded, corresponding to the processes of MeOH consumption and regeneration. The C=O signals also exhibited a similar changing trend, aligning with the process of DMC transesterification. Overall, these results clarify the specific roles of MeOH and glycerol in synergistically boosting the tandem reactions: MeOH is rapidly consumed in the PC alcoholysis step to yield BPA and DMC; glycerol then drives DMC transesterification to yield HMDL, accompanied with MeOH regeneration.

The combination of PC alcoholysis (involving MeOH) and downstream glycerol transesterification step (involving DMC) led to the formation of a closed MeOH loop, shifting the reaction equilibrium and enabling a self-driven tandem alcoholysis process. Accordingly, this process achieved rapid and nearly complete conversion of PC and glycerol under unprecedentedly mild conditions (Supplementary Video 1).³¹ The PC was fully converted within 4 hours at 80°C, yielding high-value products with high yields (96% for BPA and 90% for HMDL). This is in sharp contrast to conventional alcoholysis reactions that typically require >130 °C reaction temperature. Due to the formation of a closed-loop MeOH cycle, only a small MeOH dose is needed to initiate the tandem reactions, resulting in ultra-low chemical consumption (Fig. 1j-k). The MeOH consumption (MeOH/PC molar ratios ≤ 0.125) is far less than that of conventional PC alcoholysis processes (MeOH/PC > 5)^{15, 17, 18, 32-38}. Even taking into account the glycerol consumption, the total alcohol usage of the tandem alcoholysis process is still advantageous ((MeOH+glycerol)/PC ≤ 1.125). In addition, it also offers higher economic benefit, operational safety and process simplicity than conventional glycerol transesterification process for HMDL production, as it avoids over-stoichiometric DMC input (typically DMC: glycerol > 3) and complex downstream solvent-recovery procedures.

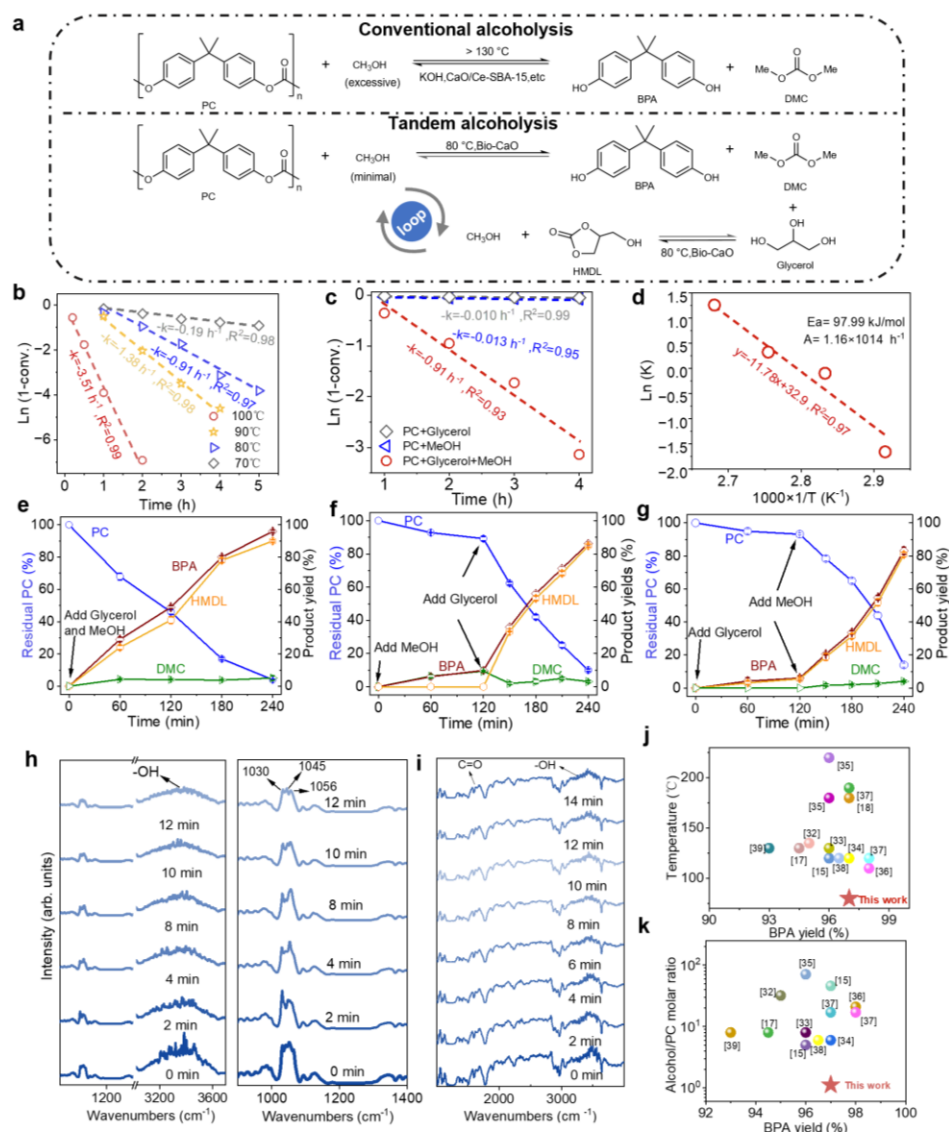


Fig. 1. Performance of the tandem alcoholysis process. **a** Comparison of tandem and conventional alcoholysis processes for PC conversion. PC conversion **b** in the presence of different alcohols and **c** at different reaction temperatures. **d** Arrhenius plots for PC tandem alcoholysis. **e–g** PC conversion and product yields over time with simultaneous addition of MeOH and glycerol, sequential addition of MeOH and glycerol, and sequential addition of glycerol and MeOH. **h–i** In situ ATR-FTIR spectra of PC depolymerization with MeOH addition, and MeOH and glycerol co-addition. **j–k** Comparison of alcohol usage and temperature in this work and the literature. Error bars of **e–g** correspond to the standard deviation of three measurements. Reaction condition for **b–g**: [catalysts] = 10 mg, [PC] = 500 mg, [glycerol] = 180 mg, [MeOH] = 70 mg.

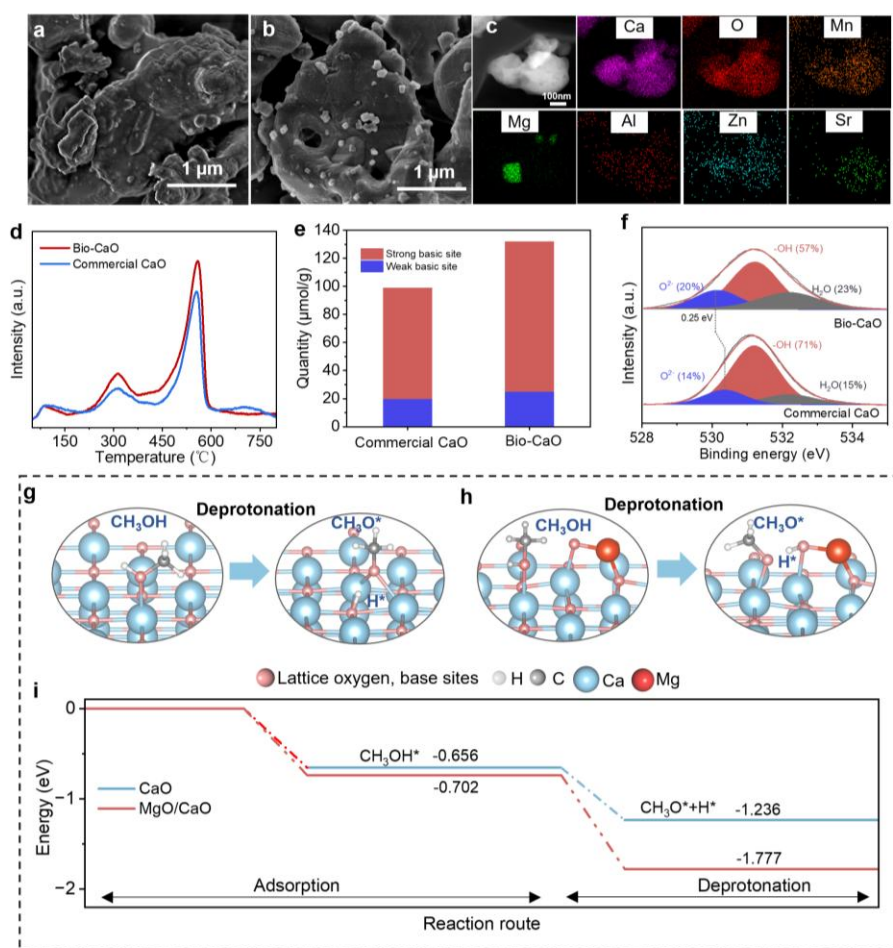
Catalytic mechanism of bio-CaO in boosting tandem alcoholysis

Apart from the rational process design, the adoption of the bio-CaO catalyst is also an important reason for the superior performance of the tandem alcoholysis system. The exceptional catalytic activity of the bio-CaO was supported by the significant performance decline when shifting to other base metal catalysts and even commercial pure CaO (Supplementary Fig. 3a). To unveil the origin of bio-CaO's high activity, we comprehensively examined its morphology, composition, and structural properties. Compositional analysis of the bio-CaO (Supplementary Table 1) identified abundant multimetal oxide nanoparticles (approximately 100 nm in size), including MgO (3.47%), ZnO (1.3%), Al₂O₃ (1.3%), MnO (0.91%), and SrO (1.4%) (Fig. 2a-c and Supplementary Fig. 7–8). These nanoparticles are uniformly dispersed on the bulk CaO surface but are absent in commercial CaO, suggesting they also contributed to the high activity of bio-CaO. The CO₂-TPD profiles (Fig. 2d-e) reveal a significantly higher density of strong basic sites on the surface of bio-CaO than on commercial CaO. Consistently, a markedly higher proportion of lattice oxygen species (O²⁻, typically serving as strong basic site) was also identified in the bio-CaO by XPS spectra (Fig. 2 f), confirming the presence of more surface O²⁻ species. Here, the metal oxide nanoparticles at the edges and corners of bulk CaO induced the formation of low-coordinated O²⁻ sites, as evidenced by the shifted O1s XPS peak toward lower binding energy. Such low-coordinated O²⁻ species further raise the surface basicity of bio-CaO^{39,40} to favor alcohol deprotonation^{41,42}. Specifically, these O²⁻ species with increased electron density could bind strongly with the alcohol hydroxyl group via acid–base interactions, strengthening the polarization of the alcohol O–H bond^{43,44}. The alkoxide intermediates (RO⁻) resulted from alcohol deprotonation would then directly attack the carbonyl carbon in both PC and DMC molecules, thereby boosting the overall tandem alcoholysis reactions.

These catalytic mechanisms were further verified by density functional theory (DFT) calculations. A simplified MgO/CaO model, with MgO supported on CaO, was established for the simulation (Fig. 2g–i, Supplementary Fig. 9). The results show that MeOH adsorption ($\Delta E_{\text{ads}} = -1.038$ eV) and deprotonation ($\Delta E_{\text{deprot}} = -1.770$ eV) at the surface O²⁻ site of MgO/CaO are thermodynamically more favorable than those on pure CaO ($\Delta E_{\text{ads}} = -0.580$ eV, $\Delta E_{\text{deprot}} = -1.236$ eV) (Fig. 2i). Therefore, the alcohol molecules can be strongly adsorbed on the bio-CaO surface to facilitate O–H bond dissociation, thus accelerating alcohol conversion relative to the CaO group. The synergistic effects between the multimetallic components and bulk CaO were also experimentally verified by synthesizing a composite catalyst MgSrMnAlZnO_x/CaO. This catalyst showed a 25% higher strong basic site density (Supplementary Fig.

10a) and a 70% higher content of coordination-unsaturated O^{2-} sites than pure CaO (Supplementary Fig. 10b), consistent with its enhanced catalytic activity.

Fig. 2. Properties and catalytic performances of bio-CaO and commercial CaO for alcohol deprotonation. a–b SEM images of commercial CaO and bio-CaO. c HAADF-STEM image and corresponding EDS mapping of the bio-CaO. d–f CO_2 -TPD profiles, distribution of strong and weak basic sites, and O 1s XPS spectra of commercial CaO and bio-CaO. g–h MeOH adsorption and deprotonation processes on CaO and MgO/CaO surfaces, and i the corresponding reaction energy barriers.



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Notably, the metal oxide nanoparticles (e.g., ZnO, Al_2O_3 , MnO) also introduced extra acid sites, which further added up to the high activity of the bio-CaO catalyst. According to the NH_3 -TPD profiles, the bio-CaO and $MgSrMnAlZnO_x/CaO$ both exhibit higher surface acidity than pure CaO (Fig. 3a–b, Supplementary Fig. 11). Typically, the metal cations such as Zn^{2+} , Al^{3+} and Mn^{2+} , owing to their high charge density and small ionic radii, can serve as strong Lewis acid sites to participate in surface reactions.⁴⁵ In our tandem alcoholysis system, these additional metal cations in bio-CaO would strongly

coordinate with the carbonyl O atoms in PC and DMC, promoting C=O bond polarization and endowing the carbonyl carbon with enhanced electrophilicity to facilitate C–O bond cleavage. Such interactions were substantiated by DFT calculations using ZnO as a model metal oxide (Supplementary Fig. 12). ZnO not only enables stronger PC adsorption than CaO ($E_{ad} = -1.40$ vs. -1.00 eV), but also can efficiently activate carbonyl carbon with larger electron transfer number (-3.954 e⁻ vs. -3.946 e⁻ for CaO) (Fig. 3c-d). Consequently, the bio-CaO provides both abundant basic and Lewis acid sites to promote alcohol deprotonation and carbonyl polarization, respectively, thereby synergistically accelerating the tandem alcoholysis reactions (Fig. 3e-f). Additionally, the notably larger specific surface area of bio-CaO (11.6 vs. 1.55 m² g⁻¹ for commercial CaO), which provides more accessible active sites, might also contributed to its superior catalytic performance (Supplementary Fig. 13).

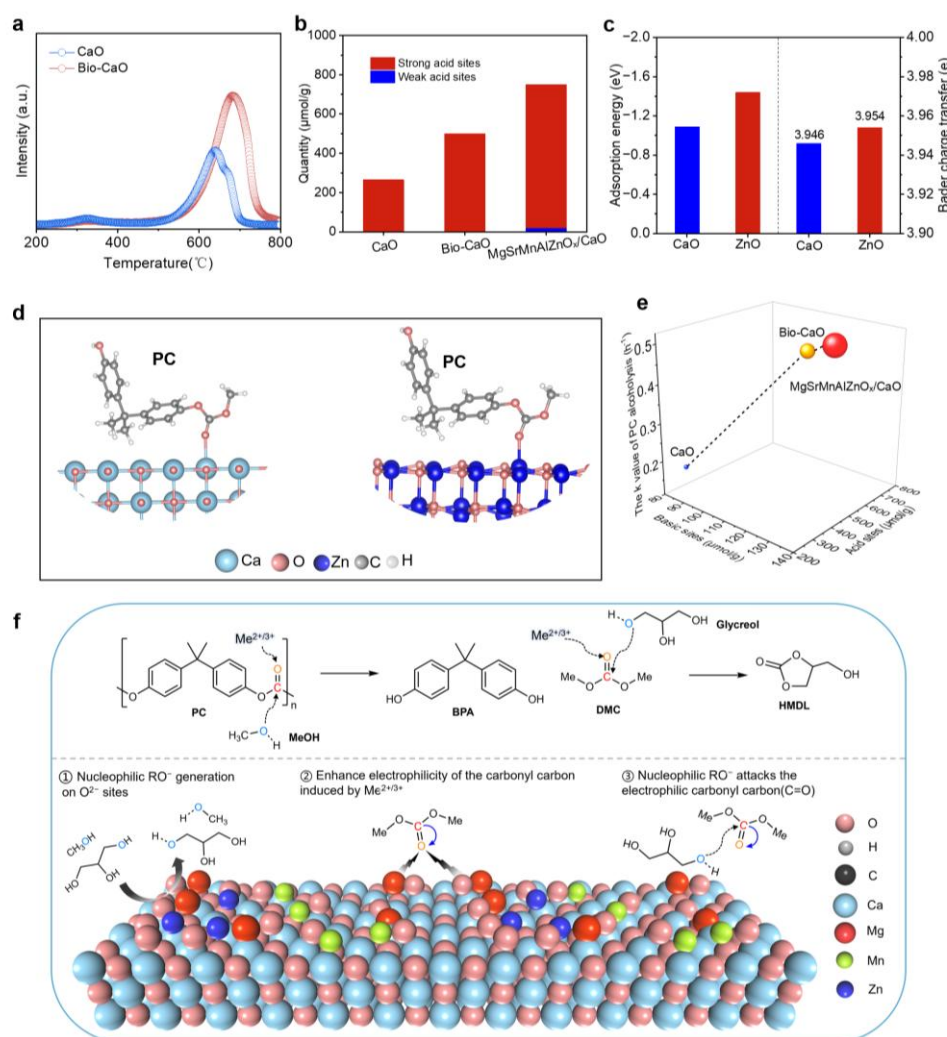


Fig. 3. Origins of the superior catalytic activity of bio-CaO in the tandem reactions. **a-b** NH₃-TPD profiles and quantitative distribution of strong and weak acid sites of commercially CaO and bio-CaO. **c** Reaction energy barrier for PC model adsorption and bader charge loss of the carbonyl carbon on ZnO and CaO. **d** Configurations of PC model adsorption on ZnO and CaO. **e** Reaction rates versus acid-base site density. Symbol size denotes glycerol transesterification rate constant. **f** Catalytic mechanisms of bio-CaO in the tandem reactions.

Performance for upcycling of real-life plastics wastes

We further evaluated the feasibility of the above tandem alcoholysis process for treating real plastics, which typically have more complicated composition with diverse impurities. An array of real-life PC plastic wastes, including glasses, compact discs (CD) and boxes, were tested under the above optimal conditions. Taking the upcycling of CD as an example (Supplementary Fig. 14), the solid waste was

rapidly converted into a clear, transparent liquid with no visible residue after the tandem alcoholysis reactions. To further separate the products, the crude liquid was recrystallized in water to precipitate BPA (solid phase), while HMDL remained in the liquid phase. Similarly, an efficient upcycling of diverse real PC plastic wastes was also achieved in this process, demonstrating 89–93% BPA and HMDL yields (Fig. 4a; Supplementary Fig. 15 and Supplementary Video 2). Considering that many real-life PC plastics also contain various additives (e.g., dyes, pigments, fillers, antioxidants, and plasticizers) that might poison the catalyst or interfere with the catalytic reactions, we also evaluated the treatment performances of several colored PCs (containing different dyes). Impressively, in all these cases the tandem alcoholysis system retained 84–89% BPA yield and 84–86% HMDL yield (Fig. 4b), demonstrating superior robustness.

Additionally, because plastic wastes typically exist as plastic blends, it is desirable to further evaluate the process adaptability to processing such mixed plastics. Here, several types of mixed plastics (all from real-life plastics) were tested, including PC-PBT (polybutylene terephthalate) copolymer, PC-ABS (Acrylonitrile butadiene styrene) copolymer, and a mixed plastics sample consisting of PC, polystyrene (PS), polyethylene terephthalate (PET), and polyvinyl chloride (PVC). Encouragingly, an efficient and selective conversion of PC was consistently achieved in all these systems regardless of the complicated feedstock compositions. For the mixed plastics group, the PS, PET, and PVC fractions were largely unreacted with only minor morphological changes, while PC was almost completely converted (Fig. 4c; Supplementary Fig. 16). Such exceptional selectivity should be mainly attributed to the mild reaction conditions (80 °C) and minimal methanol input, which prohibit the cleavage of the strong C–C backbones of PS and PVC. Additionally, PET conversion was also hindered due to the incapability of the system for rapidly consuming the PET depolymerization intermediates. Similarly, the PC+PBT and PC+ABS blends groups also witnessed rapid conversion of PC while leaving PBT or ABS as solid residue (Fig. 4d). Therefore, the tandem alcoholysis process could be applied for directly processing various PC-containing mixed-plastic streams to yield high-purity products (Fig. 4e-g and Supplementary Fig. 17–18)—a sustainable route that is unachievable by conventional high-temperature alcoholysis processes.

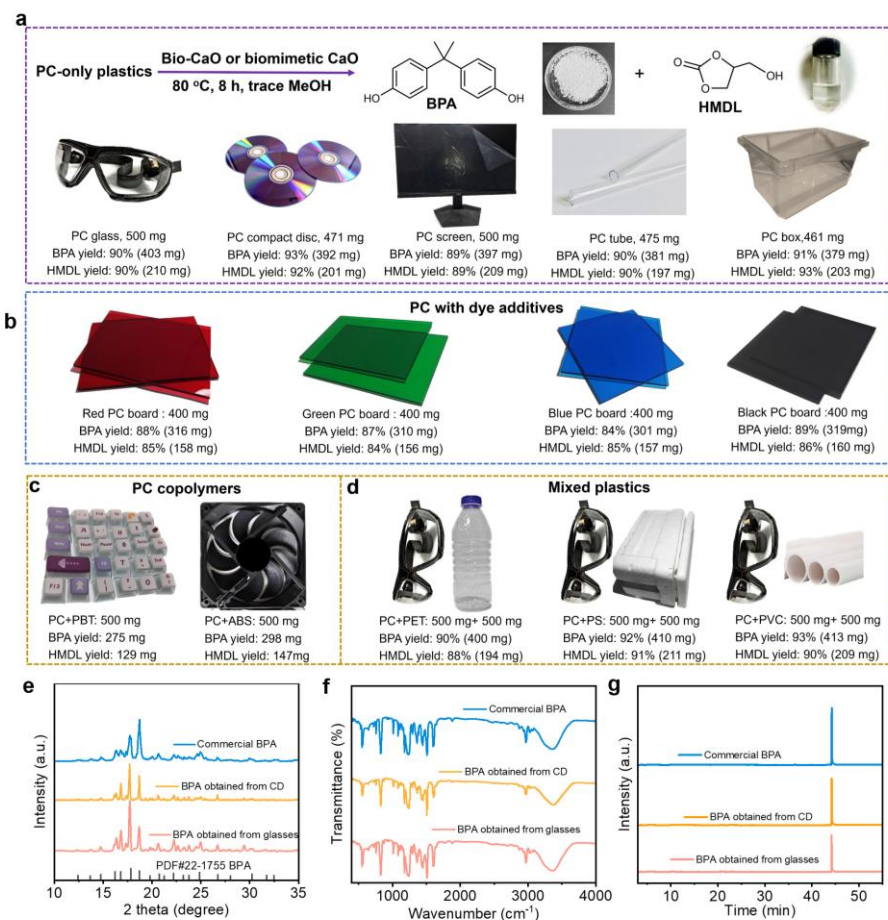


Fig. 4 Upcycling of real-life PC plastics by tandem alcoholysis process. a PC-only waste plastics. **b** PC plastics with dye additives. **c** PC copolymers. **d** Mixture of PC and other plastics. **e–g** XRD patterns, FTIR spectra, and GCMS results of BPA obtained from waste CD and glasses and the standard sample.

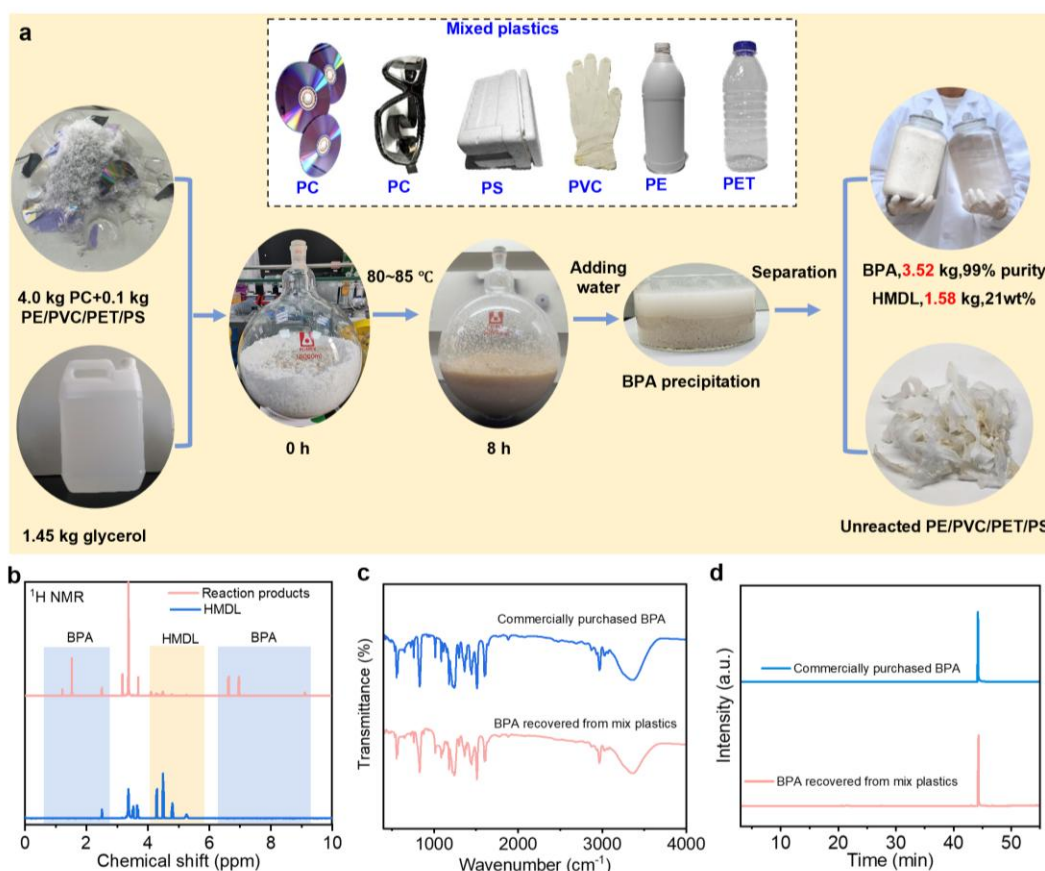
Pilot-scale demonstration

The mild operating conditions, simple process flow, robust performance and high compatibility with mixed plastic feeds endow the tandem alcoholysis system with excellent scalability and practical applicability. For demonstration, we established a pilot-scale system using a simple 10-L glass reactor vessel for processing real plastic wastes. The reactor was fed with 4.1 kg mixed plastics (consisting of PC, PET, PVC, PP, and PE) and 1.45 kg crude glycerol (Fig. 5a and Supplementary Video 3). After reaction at 80–90°C for 15 hours, the liquid products were separated. Ultimately, 3.52 kg of BPA (the solid-phase product) and 1.58 kg of HMDL (the liquid-phase product) were obtained, corresponding to a 97% BPA yield and an 86% HMDL yield. The NMR (Fig. 5b), XRD (Supplementary Fig. 19) and FTIR (Fig. 5c) characterizations show a high degree of match between the recovered products and the commercial standards, confirming high product purity. The GC-MS results (Fig. 5d) further verify a >99% purity of

the BPA product. In addition, the PET, PVC, PP, and PE fractions remained unreacted and could be readily separated from the reaction mixture.

Overall, the pilot-scale reactor steadily achieved kilogram-scale upcycling of mixed plastics with robust performance, verifying its excellent scalability. To ensure quality control and reliable supply of the catalyst to support large-scale production, we also explored the possibility of using an eggshell-inspired biomimetic catalyst as an alternative, which allows for well-controlled mass production. The fabricated $\text{MgZnMnO}_x/\text{CaO}$ exhibited catalytic performance comparable to that of bio- CaO (Supplementary Note 1.1, Supplementary Fig. 20–26 and Supplementary Table 2), further verifying the feasibility of technology upscaling.

Fig. 5. Pilot-scale demonstration for co-upcycling of PC and glycerol wastes. **a** Illustration of the



operation procedures. **b** ^1H NMR spectra of HMDL and tandem alcoholysis products. **c** FTIR spectra of the recovered BPA and standard sample. **d** GC-MS result of the recovered BPA.

Environmental impacts and economic analysis

The above results demonstrate the advantages of the tandem alcoholysis process in terms of reaction

efficiency, product quality, process robustness and scalability, and broad adaptability across a variety of real-world PC-containing plastic wastes, highlighting its technical feasibility for PC plastic upcycling. Nevertheless, environmental impacts and economic costs also matter for its practical application. Therefore, the environmental impacts of the tandem alcoholysis process for BPA and HMDL production were comprehensively evaluated by life cycle assessment (LCA), and compared with those of the conventional industrial production routes (Fig. 6a).³⁰ For the LCA, a large-scale facility with a treatment capacity of 1000 kg PC waste was assumed. The tandem alcoholysis process is enabled by adding 362.2 kg of glycerol and 16 kg of MeOH and operating at 80 °C (Fig. 6b).

The LCA results show striking environmental advantages of the tandem alcoholysis over the conventional production processes across multiple indicators (Fig. 6c, Supplementary Tables 3–10), including a 94 % lower global warming potential, 95.2% less fossil resource consumption, and approximately 95 % energy savings (Fig. 6d–f). These benefits largely stem from feedstock substitution. The conventional BPA/HMDL production processes all rely on fossil-based raw materials (phenol, acetone, DMC), and require excess reagents (e.g., phenol: acetone $\approx$ 2:1; DMC: glycerol $\approx$ 3:1). In contrast, the tandem alcoholysis route only uses waste plastics and bio-derived glycerol as the major feedstock. The overall process requires only minimal MeOH input, thus reducing the chemical consumption by $\sim$ 79%. Accordingly, this leads to $\sim$ 86% lower carbon emissions and $\sim$ 91% higher energy saving relative to the conventional production routes (Supplementary Fig. 27–29). Additionally, the lower reaction temperature, adoption of smaller-sized reactor, and avoidance of extensive recovery operations further reduce the energy consumption, contributing to $\sim$ 33%, $\sim$ 4% and $\sim$ 9% extra reduction in electricity demand, energy use, and carbon emissions, respectively. The avoided incineration of waste PC further contributes to its environmental benefits.

Lastly, the economic viability of a tandem alcoholysis facility for large-scale production of BPA and HMDL during long-term operation was estimated, assuming a waste plastic processing capacity of 150,000 tons/year. Costs and revenues were estimated based on key techno-economic parameters (Supplementary Tables 11–17). The results show that producing one ton of products (comprising 673 kg BPA and 327 kg HMDL) by the tandem alcoholysis process would cost \$554 (Supplementary Table 16). This presents a 65.7% cost saving relative to the conventional industrial process (see the Supplementary Information for details), primarily due to reduced chemical and energy consumption. The reduced cost and enhanced product value thus drastically improve its gross profit (Fig. 6g), enabling 50% higher annual net profit (\$558.9 million) than those of the industrial BPA and HMDL production processes (\$369.6 million in total). Notably, crude glycerol, as a common by-product of biomass-refining processes with

low prices ($\sim \$200/\text{ton}$)^{46, 47}, may be used as an alternative to the refined glycerol with comparable performance in the tandem process (Supplementary Note 1.2 and Supplementary Fig. 30), thus further improving the net economic revenue. Sensitivity analysis (Fig. 6h, Supplementary Fig. 31) identified the product yields and the feedstock (i.e., PC and glycerol) prices as the key factors influencing production cost. A $\pm 10\%$ fluctuation in these factors resulted in $<10\%$ changes in cost and profit, confirming its good robustness to favor sustainable and profitable chemical production.

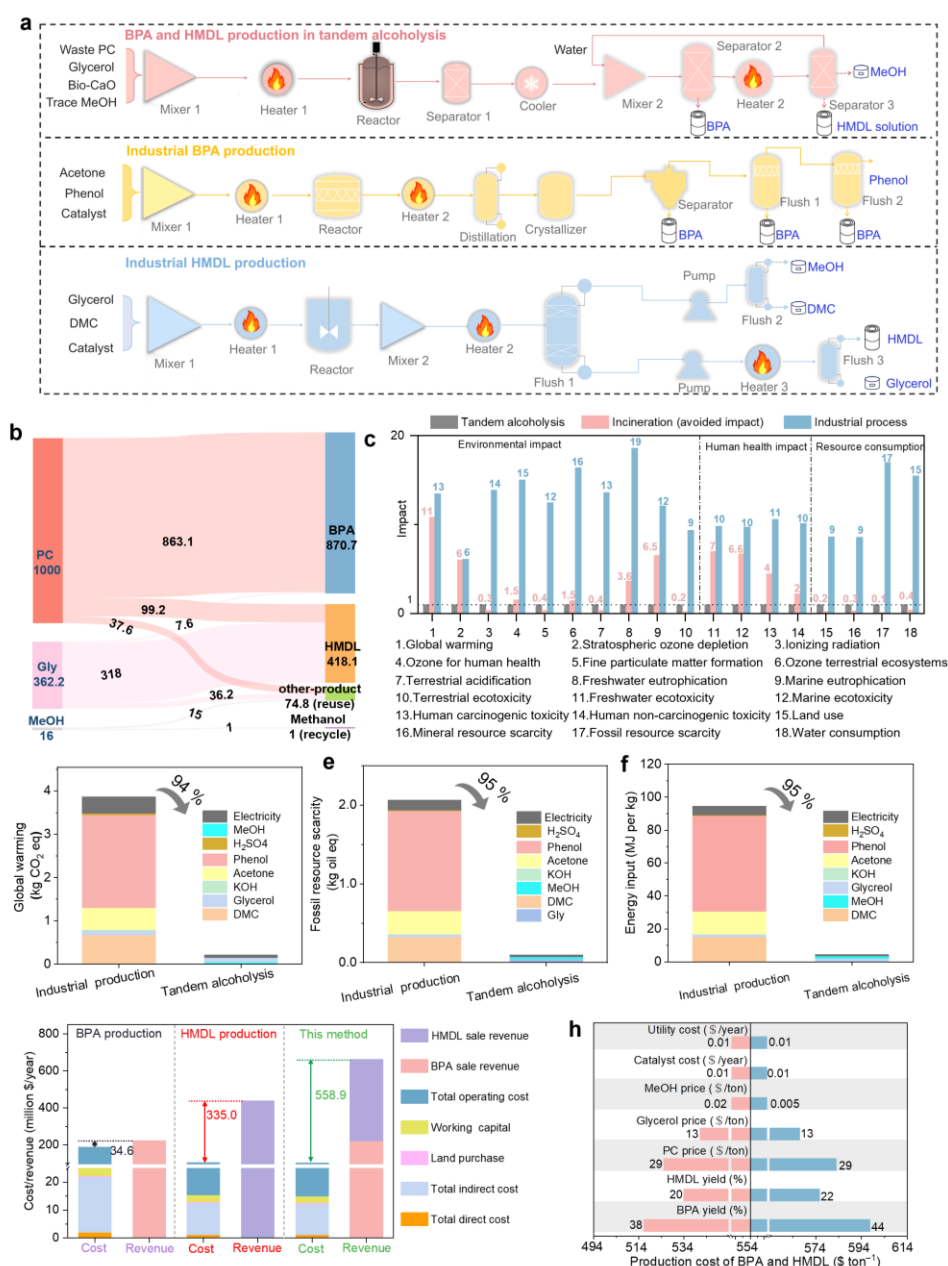


Fig. 6. Life cycle assessments and economic analysis of the tandem alcoholysis process for PC and glycerol co-upcycling. a Workflows of tandem alcoholysis process with simultaneous production of BPA

and HMDL and the conventional processes for BPA and HMDL production. **b** Sankey diagram of mass flows for the tandem alcoholysis. **c** Comparison of the environmental impacts of different processes. **d–f** Global warming potential comparison, fossil energy consumption, total energy consumption and revenues of different processes. **g** Cost and revenue comparisons of different HMDL and BPA production processes. **h** Sensitivity analysis model for production costs.

Discussion

This work reports a closed-loop, self-sustaining tandem alcoholysis platform that enables full upcycling of waste PC and glycerol into high-value products (BPA and HMDL) under mild conditions. This process couples PC alcoholysis with glycerol transesterification to form a closed-loop MeOH cycle, not only drastically lowering the chemical dosage but also breaking the reaction equilibrium to accelerate the tandem reactions. In addition, an eggshell-derived bio-CaO catalyst, with multimetallic oxide nanoparticles and abundant basic and Lewis acid sites, was adopted to facilitate an efficient and selective conversion. The overall process operated at only 80 °C (vs. >130 °C in conventional processes) with ~98% less MeOH demand, and generated BPA (96% yield) and HMDL (90% yield) as high-value products. Importantly, the mild tandem process enables selective PC conversion in mixed plastics with minimal side reactions and simplified downstream separations. Pilot-scale operation (>5 kg PC waste) in simple low-pressure reactors demonstrated its scalability and practical feasibility. Beyond plastic conversion, the process simultaneously upcycles crude glycerol—an underutilized, low-value biodiesel by-product—whereas conventional upgrading routes to HMDL typically require excess external DMC and complex separations. This dual-waste valorization further strengthens the environmental and economic advantages of our tandem process over prevailing industrial routes to BPA and HMDL. Overall, this study establishes a scalable, economically attractive, and sustainable route for waste valorization, with potential to advance waste upgrading and green chemical manufacturing.

Methods

Raw materials and catalyst preparation

The PC powder was purchased from Ruina New Materials Co., Ltd. (weight-averaged molecular mass (Mw) 33.4 kDa, number-averaged molecular mass (Mn) 19.5 kDa). The actual PC waste plastics were sourced from a nearby recycling plant and were shredded into powder before use. Waste eggshells were collected from local restaurants. Dimethyl carbonate (DMC), hydroxymethyl dioxolanone (HMDL), methanol, glycerol, CaCO_3 , $\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, $\text{Sr}(\text{NO}_3)_2$, $\text{Al}(\text{NO}_3)_3$, $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Mg}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ were procured from Sigma-Aldrich.

Eggshells were washed with tap water and distilled water to sufficiently remove the impurities on the surface. The washed eggshells were then dried in a hot air oven at 105 °C, ground with a grinder, and sieved through a 60-mesh sieve. Next, the eggshell powder was calcined in a muffle furnace (in air) at 900 °C for 4 hours to obtain bio-CaO. The as-synthesized bio-CaO powder was stored in the close vessel before use. $\text{MgSrMnAlZnO}_x/\text{CaO}$, $\text{MgMnZnO}_x/\text{CaO}$, SrO_x/CaO , and ZnO_x/CaO were prepared via aqueous impregnation. Metal loading was determined from the elemental composition of the eggshell (Supplementary Table 1). CaCO_3 was dispersed in deionized water and then loaded with stoichiometric metal nitrates $\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, $\text{Sr}(\text{NO}_3)_2$, $\text{Al}(\text{NO}_3)_3$, $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Mg}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ by stirring at room temperature for 12 h. The resulting mixture was then dried at 110 °C and subsequently calcined in air at 900 °C for 4 h. As a control, pure CaO was obtained by directly calcining CaCO_3 under identical conditions.

Catalytic evaluation

Tandem alcoholysis reactions were conducted in 2-20 mL glass vessels, which was placed in oil bath equipped with a K-type thermocouple for temperature control. Bio-CaO and PC were finely ground and uniformly mixed, followed by addition of MeOH and glycerol. The mixture was heated under magnetic stirring, and at the end of each reaction, samples were collected for quantitative analysis using gas chromatography-mass spectrometry (GC-MS), high-performance liquid chromatography (HPLC) and proton nuclear magnetic resonance (NMR). The PC and DMC conversion ratios and the BPA and HMDL yields were calculated by the following equations:

$$\text{PC conversion}(\%) = \frac{W_{\text{pc,i}} - W_{\text{pc,f}}}{W_{\text{pc,i}}} \times 100\% \quad (1)$$

$$\text{Yield}(\%) = \frac{W_{\text{p,e}}}{W_{\text{p,t}}} \times 100\% \quad (2)$$

where $W_{\text{pc,i}}$ is the initial weight of the PC, $W_{\text{pc,f}}$ is the final weight of the PC, $W_{\text{p,e}}$ is the weight

of the product obtained in the experiment, and $W_{p,t}$ is the theoretical yield of the product.

A pilot-scale facility of the tandem alcoholysis process was established using a simple 10-L glass vessel as the reactor. Firstly, 4.1 kg of mixed plastic wastes, including 4.0 kg of PC and 0.1 kg of other common thermoplastics (i.e., PET, PVC, PP, and PE) and 80g bio-CaO were added in the reactor. Additionally, 1.45 kg glycerol and 64 g MeOH (MeOH/PC molar ratio=0.125) were also added and sufficiently mixed with the solid feedstocks. Next, the tandem alcoholysis reaction was initiated by heating the mixture to 90 °C within 1 hour. Then the temperature was held at 80–85 °C for 15 hours to ensure sufficient conversion of PC and glycerol.

Characterization of catalysts and products

The catalyst composition and alcoholysis products were analyzed by XRD with 2θ range of 10–80° (miniflex600, Rigaku). XRF (S4 PIONEER, BRUKER) and ICP-MS (NexION 5000) were used to determine the element contents of bio-CaO. The element valances were determined by XPS spectroscopy (Thermo Fisher Scientific K-Alpha, USA). The material morphology and composition were examined by SEM (GeminiSEM 450, Zeiss) and HRTEM (Talos F200X). The acidity and basicity of the catalysts were measured by CO₂-TPD and NH₃-TPD analysis (Micromeritics AutoChem II 2920). N₂-sorption measurements were conducted at 77 K with a Micromeritics Tristar 3000 instrument to determine the BET surface area. The functional group changes of the reactants were monitored using a FT-IR spectrometer (Nicolet 5700, Thermo Fisher Scientific).

Product analysis

The BPA and HMDL products in the liquid phase were identified and quantified by GC-MS (Agilent 7890B-5977B) equipped with an HP-5MS column (30 m × 0.25 mm). Split injection was performed at a split ratio of 20 :1 using argon (99.99%) as carrier gas. The column heating procedure was as follows: (1) holding at 40 °C for 1 min, (2) heating to 180 °C at 4 °C/min and holding at 180 °C for 2 min, and then (3) heating to 250 °C at 5 °C/min and holding at 250 °C for 2 min. HMDL was collected and diluted with deionized water for analysis with high performance liquid chromatography (HPLC, LC-16, Shimadzu Co., Japan) equipped with a RID detector. After reaction, the mixture was diluted with deionized water and subsequently filtered to remove any solid residues. Subsequently, 10 µl solution was injected into HPLC for analysis. The column was operated at 50 °C and the mobile phases of HPLC in this method were 10% methanol and 90% sulfuric acid solution (5 mM, flow rate 0.6 mL/min).

Theoretical calculations

All first-principles calculations were performed with the Vienna Ab-initio Simulation Package (VASP,5.4.1) using density functional theory within the generalized-gradient approximation (GGA). The Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional was employed together with the projector-augmented-wave (PAW) approach for the electron–ion interactions. The plane wave cutoff energy was set as 450 eV. Long-range dispersion was included via Grimme’s DFT-D3 correction. Structural optimization was carried out with energy convergence criteria set at 1.0×10^{-5} eV and force convergence criteria set at $0.05 \text{ eV } \text{\AA}^{-1}$. The Brillouin zone was sampled with a Γ -centered $(2 \times 2 \times 1)$ k-point mesh for geometry relaxations; charge analyses (Bader) were then carried out on densities obtained with a $3 \times 3 \times 1$ Monkhorst–Pack grid. Gibbs free-energy changes (ΔG) for the elementary steps are evaluated using the following formula:

$$\Delta G = \Delta E + \Delta E_{\text{ZPE}} - T\Delta S \quad (3)$$

where ΔE is the electronic energy difference directly obtained from DFT calculations, ΔE_{ZPE} is the zero-point-energy difference, and ΔS is the entropy change; unless otherwise noted, T is the room temperature (353.15 K).

Comprehensive Life-Cycle Assessment (LCA)

The LCA of the tandem alcoholysis processes and the industrial processes for HMDL and BPA production were performed using the SimaPro 9.5 software. The results were categorized into 18 environmental impact indicators using the ReCiPe 2016 Midpoint (H) method. The life cycle inventory analysis considered the input/output data regarding the system (include the material quantities and electricity). The “cut-off” approach was applied to waste PC, assigning no upstream environmental impact for choosing waste as the feedstock. The environmental impact categories include environmental impacts, resource consumptions, and human health impacts. Here, the ecosystem impacts primarily include global warming potential, stratospheric ozone depletion, ionizing radiation, ozone formation, and marine eutrophication. Resource consumption impacts mainly involve fossil fuel depletion, mineral resource usage, water consumption, and land use. The human health impacts include terrestrial ecotoxicity, freshwater ecotoxicity, marine ecotoxicity, human carcinogenic toxicity, and human non-carcinogenic toxicity (more details see Supplementary Note 1.3).

Model development and techno-economic analysis

The process models in this study were developed according to experimental results using Aspen Plus

V12 modeling environment. The poly-NRTL property method is chosen as a physical properties package for tandem alcoholysis. The techno-economic analysis was conducted based on Aspen simulation results, with the designed plant has an annual processing capacity of 150,000 tons of waste PC. The analysis includes a comprehensive evaluation of fixed investments, energy requirements, raw material costs, and production sales (more details see Supplementary Note 1.4).

In situ FT-IR study

The solid samples collected during PC alcoholysis and tandem alcoholysis were subjected to in situ FT-IR analysis using an infrared beamline BL01B of the National Synchrotron Radiation Laboratory (NSRL, Hefei, China). Experiments on PC alcoholysis and tandem alcoholysis catalyzed by bio-CaO was conducted in a sealed IR cell. The scans were acquired in the wavenumber range of 4000–600 cm^{-1} . The averages of 64 individual scans were taken as the final data (more details see Supplementary Note 1.5).

Data availability

The source data generated in this study are available in the figshare repository (<https://figshare.com/account/articles/31771216?file=62865091>). Additional data are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

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Author contributions

Y.K., W.J.L., and W.W.L. devised the concept, Y.K. conducted the main experiments. W.J.L., W.W.L. and H.Q.Y. supervised the project. Y.Z.C., Z.Y.G., and S.X.W. helped for the materials characterizations and products analysis. Y.K. and Z.Y.G. wrote the manuscript. W.J.L. and W.W.L. revised the manuscript.

Competing interests: The authors declare no competing interests

Figure Legends/Captions

Fig. 1. Performance of the tandem alcoholysis process. **a** Comparison of tandem and conventional alcoholysis processes for PC conversion. **b** PC conversion in the presence of different alcohols and **c** at different reaction temperatures. **d** Arrhenius plots for PC tandem alcoholysis. **e–g** PC conversion and product yields over time with simultaneous addition of MeOH and glycerol, sequential addition of MeOH and glycerol, and sequential addition of glycerol and MeOH. **h–i** In situ ATR–FTIR spectra of PC depolymerization with MeOH addition, and MeOH and glycerol co-addition. **j–k** Comparison of alcohol usage and temperature in this work and the literature. Error bars of **e–g** correspond to the standard deviation of three measurements. Reaction condition for **b–g**: [catalysts] = 10 mg, [PC] = 500 mg, [glycerol] = 180 mg, [MeOH] = 70 mg.

Fig. 2. Properties and catalytic performances of bio-CaO and commercial CaO for alcohol deprotonation. **a–b** SEM images of commercial CaO and bio-CaO. **c** HAADF-STEM image and corresponding EDS mapping of the bio-CaO. **d–f** CO₂-TPD profiles, distribution of strong and weak basic sites, and O 1s XPS spectra of commercial CaO and bio-CaO. **g–h** MeOH adsorption and deprotonation processes on CaO and MgO/CaO surfaces, and **i** the corresponding reaction energy barriers.

Fig. 3. Origins of the superior catalytic activity of bio-CaO in the tandem reactions. **a–b** NH₃-TPD profiles and quantitative distribution of strong and weak acid sites of commercially CaO and bio-CaO. **c** Reaction energy barrier for PC model adsorption and bader charge loss of the carbonyl carbon on ZnO and CaO. **d** Configurations of PC model adsorption on ZnO and CaO. **e** Reaction rates versus acid-base site density. Symbol size denotes glycerol transesterification rate constant. **f** Catalytic mechanisms of bio-CaO in the tandem reactions.

Fig. 4 Upcycling of real-life PC plastics by tandem alcoholysis process. **a** PC-only waste plastics. **b** PC plastics with dye additives. **c** PC copolymers. **d** Mixture of PC and other plastics. **e–g** XRD patterns, FTIR spectra, and GCMS results of BPA obtained from waste CD and glasses and the standard sample.

Fig. 5. Pilot-scale demonstration for co-upcycling of PC and glycerol wastes. **a** Illustration of the operation procedures. **b** ¹H NMR spectra of HMDL and tandem alcoholysis products. **c** FTIR spectra of the recovered BPA and standard sample. **d** GC-MS result of the recovered BPA.

Fig. 6. Life cycle assessments and economic analysis of the tandem alcoholysis process for PC and glycerol co-upcycling. **a** Workflows of tandem alcoholysis process with simultaneous production of BPA

and HMDL and the conventional processes for BPA and HMDL production. **b** Sankey diagram of mass flows for the tandem alcoholysis. **c** Comparison of the environmental impacts of different processes. **d–f** Global warming potential comparison, fossil energy consumption, total energy consumption and revenues of different processes. **g** Cost and revenue comparisons of different HMDL and BPA production processes. **h** Sensitivity analysis model for production costs.

Editorial summary:

Recycling polycarbonate plastics is energy intensive and depends on excess methanol. This work couples plastic breakdown with glycerol conversion to regenerate methanol in situ, reducing energy use while achieving high yields at pilot scale.

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